

A Course in Modern Macroeconomics

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Complete Solutions

Chapter 5 — Theory and Evidence

All nine end-of-chapter problems, 5.1–5.9 (pp. 93–99)

Every problem carries the full question text, a step-by-step solution with no skipped algebra, an economic reading of the result, and a verification. Numerical answers are reproduced by the companion scripts in `Resolucao/kurlat_ch05_codigo/`, which check each analytical formula against an independent computation and abort on any mismatch.

Problems are grouped by **topic**, not by number, following the order of the chapter's own sections. Blue boxes are odd-numbered problems, purple boxes even.

Kurlat publishes no answer key, so every result here is verified by other means: executed numerical checks (marked ✓), internal consistency (the three GDP methods are made to agree; steady states are substituted back), limiting cases, and cross-book comparison with Romer (2012) ch. 1 and Jones (2020) chs. 5–6, both in `Livros/`.

This is a companion to `kurlat_solutions_ch1-4.tex` and `kurlat_solutions_ch09.tex`; chapters 6–7 follow, and chapter 8 is outside the course syllabus.

Contents

1 Chapter 5 Theory and Evidence	2
1.1 Calibration and the Speed of Convergence	3
1.2 Growth Accounting and the Sources of Growth	8
1.3 National Accounts, Factor Shares and the Golden Rule	11
1.4 Mismeasurement as a Source of Measured TFP	18
Companion code	27

Chapter 5 Theory and Evidence

Chapter 4 built the Solow model. Chapter 5 asks whether it survives contact with data, and the whole chapter is organised around one move: **put numbers on the model, then see what breaks.**

The calibration (§5.2). $\alpha = 0.35$ from the labour share; $g = 0.015$ from US GDP per capita since 1800; $n = 0.01$ from US population since 1950; $\delta = 0.04$; $s = 0.2$ from the investment rate. These five numbers are used again and again below, and they are mutually consistent: they imply

$$\frac{K}{Y} = \frac{s}{\delta + n + g} = 3.08, \quad r = \alpha \frac{\delta + n + g}{s} - \delta = 7.4\%,$$

the first of which matches the data and the second of which does not (observed safe real rates are 1–2%). Holding both facts in view is the point of the section.

The capital-accumulation hypothesis (§5.3). If countries differed only in K/L , poor countries would grow faster. Differentiating the transition gives

$$g_y = s f'(k_t) - \alpha \delta, \quad (5.3.1)$$

which is decreasing in k_t : *convergence*. The data (Fig. 3.3.1) show no such pattern, so the conjecture fails and something other than capital must differ.

Growth accounting (§5.4). A first-order Taylor expansion of $Y = F(K, L, A)$ gives

$$g_Y = \underbrace{\alpha}_{\text{capital share}} g_K + \underbrace{(1 - \alpha)}_{\text{labour share}} g_L + \underbrace{\text{Solow residual}}_{\text{TFP}}. \quad (5.4.2)$$

The key insight is that *factor shares are the right weights*, because in a competitive economy each share equals the corresponding output elasticity. Everything on the right is measurable except the last term, which is therefore obtained as a residual.

Where TFP differences come from (§5.5). Human capital, institutions and misallocation (Hsieh and Klenow, 2009). The exercises hammer one particular channel: *a measurement error in an input shows up as a difference in measured TFP*. Problems 5.4, 5.5 and 5.9 are three versions of the same trap, and they are the reason a residual should never be read as “technology” without argument.

Calibration and the Speed of Convergence

Question

Suppose the economy is described by the Solow growth model with the parameter values we used in Section 5.2.

- (a) Suppose the economy starts out with a level of GDP per efficiency unit of labor that is only 10% of its steady state level, i.e. $\tilde{y}_0 = 0.1 \tilde{y}_{ss}$. What is the initial capital stock $\tilde{k}_0$? Use a spreadsheet to compute $\tilde{k}_t$ and $\tilde{y}_t$ for $t = 1, 2, \dots, 100$. How many years does it take for $\tilde{y}_t/\tilde{y}_{ss} > 0.95$?
- (b) Suppose the economy begins in steady state but there is a fall in the rate of population growth from $n = 0.01$ to $n = 0$. How much higher will GDP per efficiency unit of labor be in the new steady state? Use a spreadsheet to compute $\tilde{k}_t$ and $\tilde{y}_t$ for $t = 1, 2, \dots, 100$. How many years does it take for $\tilde{y}_t/\tilde{y}_{ss} > 0.95$?

Setup. The calibration of §5.2 is

$$\alpha = 0.35, \quad s = 0.20, \quad \delta = 0.04, \quad n = 0.01, \quad g = 0.015,$$

with $\tilde{y} = f(\tilde{k}) = \tilde{k}^\alpha$ and the law of motion (4.5.3)

$$\tilde{k}_{t+1} = \tilde{k}_t + \frac{s \tilde{k}_t^\alpha - (\delta + n + g) \tilde{k}_t}{1 + n + g}. \quad (*)$$

Setting $\Delta \tilde{k} = 0$ gives the steady state

$$\tilde{k}^* = \left(\frac{s}{\delta + n + g} \right)^{\frac{1}{1-\alpha}}, \quad \tilde{y}^* = \left(\frac{s}{\delta + n + g} \right)^{\frac{\alpha}{1-\alpha}}.$$

With $\delta + n + g = 0.065$ and $s/(\delta + n + g) = 3.0769$,

$$\tilde{k}^* = 3.0769^{1/0.65} = \boxed{5.6357}, \quad \tilde{y}^* = 3.0769^{0.35/0.65} = \boxed{1.8316},$$

and $\tilde{k}^*/\tilde{y}^* = 3.0769 = s/(\delta + n + g)$, which is exactly (5.2.1). ✓

Solution:

- (a) **Starting 90% below.** Output, not capital, is at 10% of its steady-state value, so invert the production function:

$$\tilde{y}_0 = 0.1 \tilde{y}^* \iff \tilde{k}_0^\alpha = 0.1 (\tilde{k}^*)^\alpha \iff \tilde{k}_0 = 0.1^{1/\alpha} \tilde{k}^*.$$

With $\alpha = 0.35$, $0.1^{1/0.35} = 0.1^{2.857} = 1.3895 \times 10^{-3}$, so

$$\tilde{k}_0 = 1.3895 \times 10^{-3} \times 5.6357 = \boxed{0.007831},$$

which is only 0.139% of $\tilde{k}^*$.

This is the first thing to notice, and it is not a rounding artefact. Because α is small, output is very insensitive to capital: to lose 90% of output you must lose 99.86% of the capital stock. Countries that are ten times poorer than the US are nowhere near 1/700 of its capital per worker, which is the same arithmetic that kills the capital-accumulation hypothesis in §5.3.

Iterating (*) from $\tilde{k}_0$:

t	$\tilde{k}_t$	$\tilde{y}_t/\tilde{y}^*$
0	0.0078	0.1000
10	1.0148	0.5488
25	2.8355	0.7863
50	4.5796	0.9299
58	4.8733	0.9504
75	5.2579	0.9760
100	5.5027	0.9917

The ratio first exceeds 0.95 at

$$t = 58 \text{ years}.$$

✓ (computed in `ch05_numerico.py`, and unchanged if one uses the exact law of motion instead of $(*)$ — see the trap box below).

- (b) **A fall in population growth.** Only $\delta + n + g$ changes, from 0.065 to 0.055. Since $\tilde{y}^* \propto (s/(\delta + n + g))^{\alpha/(1-\alpha)}$,

$$\frac{\tilde{y}_{\text{new}}^*}{\tilde{y}_{\text{old}}^*} = \left(\frac{\delta + n_{\text{old}} + g}{\delta + n_{\text{new}} + g} \right)^{\frac{\alpha}{1-\alpha}} = \left(\frac{0.065}{0.055} \right)^{0.5385} = 1.0941,$$

so GDP per efficiency unit of labour is

$$9.41\% \text{ higher}$$

in the new steady state, with $\tilde{k}^*$ rising from 5.6357 to 7.2873 and $\tilde{y}^*$ from 1.8316 to 2.0040. ✓

Iterating $(*)$ with $n = 0$ from $\tilde{k}_0 = 5.6357$:

t	$\tilde{k}_t$	$\tilde{y}_t/\tilde{y}_{\text{new}}^*$
0	5.6357	0.9140
5	5.8961	0.9285
10	6.1171	0.9406
15	6.3040	0.9505
20	6.4618	0.9588
40	6.8798	0.9801
100	7.2394	0.9977

The ratio exceeds 0.95 at

$$t = 15 \text{ years}.$$

✓

Reading. The two parts look like the same experiment run twice, and the contrast between 58 years and 15 years is the lesson. Convergence in the Solow model happens at

a roughly constant *proportional* rate — Problem 5.2 shows it is $(1 - \alpha)(\delta + n + g) \approx 3.6\%$ per year here — so the time it takes is governed almost entirely by *how far the economy starts from its target*, not by what moved it. Part (a) starts at 10% of the target and needs six decades; part (b) starts at 91% of the target and needs fifteen years.

Two further points on part (b) that are easy to miss:

- Slower population growth raises the *level* of GDP per person by 9.41% but leaves its long-run *growth rate* at $g = 1.5\%$. Level effects and growth effects are different objects, and only technological progress does the second.
- Total GDP grows more slowly afterwards, at g instead of $n + g$. A country that likes the level effect must accept the aggregate one; which of the two matters depends on whether the question is about living standards or about the size of the economy.

Where this goes wrong

Kurlat's (4.5.3) drops the ng cross-term, so $(*)$ is an approximation to the exact recursion

$$\tilde{k}_{t+1} = \frac{(1 - \delta)\tilde{k}_t + s\tilde{k}_t^\alpha}{(1 + n)(1 + g)}.$$

It is worth checking that the answers do not depend on which one you use. With the exact version, $\tilde{k}^* = 5.6158$ instead of 5.6357, the steady-state gain in part (b) is 9.55% instead of 9.41%, and the two answers become 58 years and 16 years. The qualitative conclusion is identical and the quantitative one moves by at most one year; `problema_5_1_exato()` runs this check.

The genuine error to avoid is setting $\tilde{k}_0 = 0.1\tilde{k}^*$. That is a 90% cut in *capital*, which leaves output at $0.1^{0.35} = 45\%$ of steady state, not 10%, and the answer to (a) then comes out around 34 years.

5.1 — convergence under the calibration of §5.2 ($\alpha = 0.35$, $s = 0.2$, $\delta = 0.04$, $g = 0.015$)

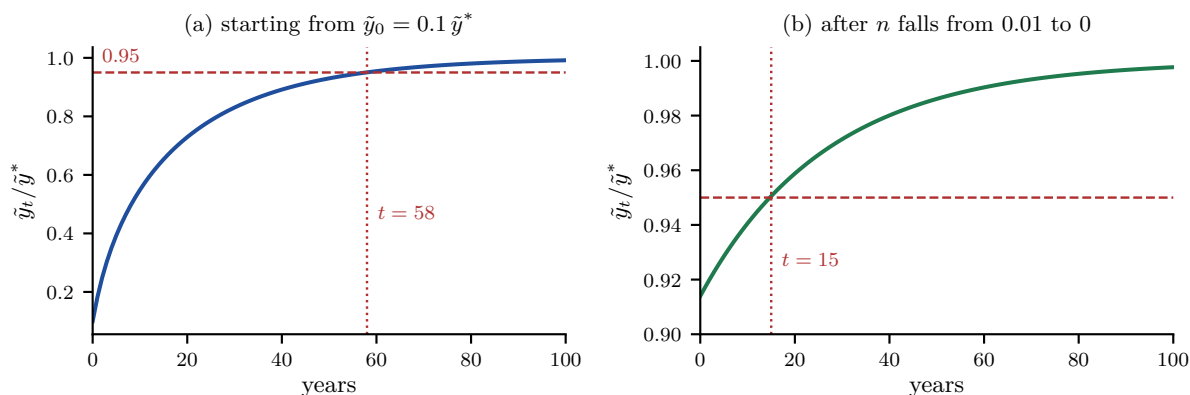


Figure 1: Problem 5.1. Left: the transition from $\tilde{y}_0 = 0.1\tilde{y}^*$. Right: the transition after n falls to zero. Same model, same speed — different starting distance.

Problem 5.2 The Speed of Convergence

Question

Suppose the economy is described by the Solow growth model with a Cobb-Douglas production function, no population growth and no technological progress. The economy is not in steady state. GDP per capita in period t is $y_t = \phi y_{ss}$, with $\phi \in (0, 1)$.

- (a) How far away from steady state is the capital stock? Find an expression for the level of k_t/k_{ss} that would be consistent with $y_t = \phi y_{ss}$, in terms of ϕ and parameters.
- (b) Find an expression for k_{ss} , the steady state level of capital, in terms of parameters.
- (c) Use your results from parts (a) and (b) and equation (5.3.1) to find an expression for g_y , the rate of growth of GDP, as a function of ϕ and parameters.
- (d) Let $\alpha = 0.35$ and $\delta = 0.04$. Compute g_y for an economy whose GDP is $\phi = 0.8$ times its steady-state level. Compute $g_y/(1 - \phi)$. What fraction of the gap between y_t and y_{ss} does the economy close in one year? Repeat this calculation for $\phi = 0.9$ and $\phi = 0.99$.

Solution:

- (a) Output is $y = k^\alpha$, so

$$\phi = \frac{y_t}{y_{ss}} = \left(\frac{k_t}{k_{ss}} \right)^\alpha \quad \implies \quad \boxed{\frac{k_t}{k_{ss}} = \phi^{1/\alpha}}.$$

Since $1/\alpha = 2.86 > 1$ and $\phi < 1$, capital is proportionally *much* further from its steady state than output is: at $\phi = 0.8$, $k_t/k_{ss} = 0.8^{2.857} = 0.53$.

- (b) With $n = g = 0$ the law of motion is $\Delta k = sk^\alpha - \delta k$, so $sk_{ss}^\alpha = \delta k_{ss}$ and

$$\boxed{k_{ss} = \left(\frac{s}{\delta} \right)^{\frac{1}{1-\alpha}}}, \quad \text{equivalently} \quad \frac{k_{ss}^{\alpha-1}}{1} = \frac{\delta}{s}. \quad (5.2.b)$$

The second form is the one that will be used — it says $s k_{ss}^{\alpha-1} = \delta$.

- (c) Equation (5.3.1) for a Cobb-Douglas $f(k) = k^\alpha$ reads

$$g_y = s f'(k_t) - \alpha \delta = s \alpha k_t^{\alpha-1} - \alpha \delta.$$

Substitute $k_t = \phi^{1/\alpha} k_{ss}$:

$$k_t^{\alpha-1} = \phi^{\frac{\alpha-1}{\alpha}} k_{ss}^{\alpha-1} = \phi^{-\frac{1-\alpha}{\alpha}} \cdot \frac{\delta}{s},$$

where the last step uses (5.2.b). Therefore

$$g_y = s \alpha \cdot \phi^{-\frac{1-\alpha}{\alpha}} \frac{\delta}{s} - \alpha \delta \quad \implies \quad \boxed{g_y(\phi) = \alpha \delta \left(\phi^{-\frac{1-\alpha}{\alpha}} - 1 \right)}$$

Note that s has cancelled: given how far output is from *its own* steady state, the growth rate does not depend on the saving rate. Two checks: $g_y(1) = 0$ ✓, and g_y is decreasing in ϕ ✓, which is convergence.

(d) With $\alpha\delta = 0.014$ and $(1 - \alpha)/\alpha = 1.8571$:

ϕ	$\phi^{-1.8571}$	g_y	$g_y/(1 - \phi)$ (as asked)	$g_y\phi/(1 - \phi)$ (gap actually closed)
0.80	1.5134	0.7189%	3.5943%	2.8755%
0.90	1.2161	0.3026%	3.0257%	2.7232%
0.99	1.0188	0.0264%	2.6376%	2.6113%

✓ Each row is confirmed in `ch05_numerico.py` against one exact iteration of $k_{t+1} = k_t + sk_t^\alpha - \delta k_t$, which reproduces g_y to within 0.7% of its own value — the small discrepancy is the first-order approximation inside (5.3.1), not an algebra error.

Which column answers “what fraction of the gap is closed”? The gap is $y_{ss} - y_t = (1 - \phi)y_{ss}$, and one year of growth adds $g_y y_t = g_y\phi y_{ss}$, so the fraction closed is $g_y\phi/(1 - \phi)$ — the last column. The exercise asks for $g_y/(1 - \phi)$, which is the same object divided by ϕ and therefore slightly larger; the two agree to three digits once ϕ is near one, which is the case the exercise is really about.

Reading, and this is the number to remember. Take $\phi \rightarrow 1$ and expand $\phi^{-(1-\alpha)/\alpha} \approx 1 + \frac{1-\alpha}{\alpha}(1 - \phi)$:

$$g_y \approx \alpha\delta \cdot \frac{1-\alpha}{\alpha}(1 - \phi) = \underbrace{\delta(1-\alpha)}_{\lambda}(1 - \phi).$$

So the economy closes a constant fraction

$$\lambda = \delta(1 - \alpha) = 0.04 \times 0.65 = \boxed{2.6\%}$$

of its remaining gap every year, and the table converges to exactly that: 3.59%, 3.03%, 2.64%. In the general case with population growth and technological progress the same derivation gives $\lambda = (1 - \alpha)(\delta + n + g)$, which under the §5.2 calibration is $0.65 \times 0.065 = 4.2\%$.

The implied **half-life** is $\ln 2/\lambda = 0.693/0.026 \approx 27$ years here (about 16 years under the full calibration). This is the single most-quoted quantitative prediction of the Solow model, and it is the number that empirical growth regressions — the “2% convergence” literature — were built to test. That the model delivers roughly the right speed while failing the level test of §5.3 is the tension the chapter is about.

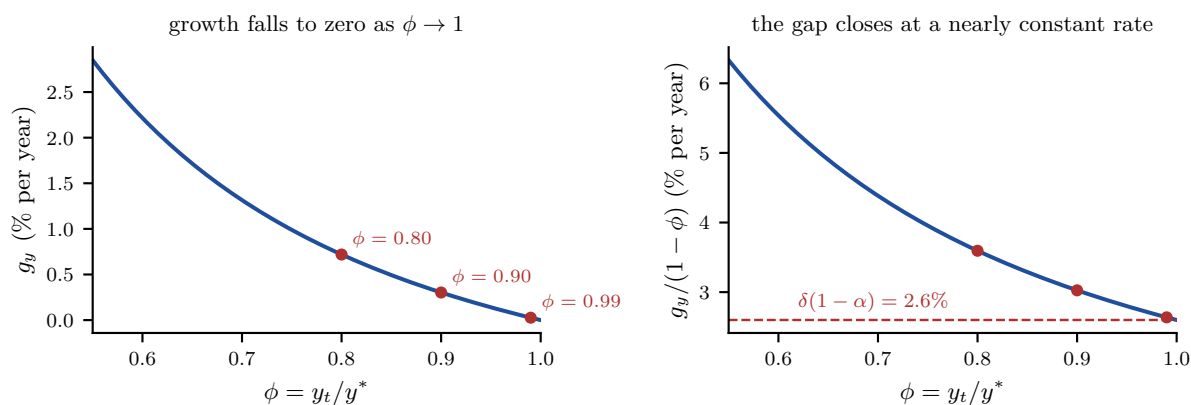
5.2(d) — speed of convergence, $\alpha = 0.35$, $\delta = 0.04$ 

Figure 2: Problem 5.2(d). Left: growth vanishes as the economy approaches its steady state. Right: the *proportional* speed at which the gap closes is nearly constant, and tends to $\delta(1-\alpha)$.

Growth Accounting and the Sources of Growth

Problem 5.3 Causes of Growth and Growth Accounting

Question

Suppose an economy is well described by the steady state of the Solow Growth Model with constant technological progress and no population growth. Imagine we take data from this economy and do a growth-accounting exercise. How much growth will we attribute to capital accumulation and how much to technology? How does this relate to the result that says that without technological progress there would be no growth in the steady state?

Solution. Write the production function in the labour-augmenting form $Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$ with $g_A = g$, $g_L = 0$.

Step 1: what the variables actually do in this steady state. By definition of a steady state, $\tilde{k} = K/(AL)$ is constant, so K grows at the same rate as AL :

$$g_K = g_A + g_L = g.$$

And $Y = \tilde{k}^\alpha AL$ with $\tilde{k}$ constant gives $g_Y = g$ as well. Everything grows at g ; this is the balanced growth path.

Step 2: run (5.4.2). The capital share is α , the labour share $1 - \alpha$:

$$\underbrace{g_Y}_g = \underbrace{\alpha g_K}_{\alpha g} + \underbrace{(1 - \alpha)g_L}_0 + \underbrace{\text{residual}}_? \implies \text{residual} = g - \alpha g = (1 - \alpha)g.$$

So the exercise attributes

a fraction $\alpha = 35\%$ of growth to capital accumulation

and a fraction $1 - \alpha = 65\%$ to technology.

Under the §5.2 calibration, of the 1.5% annual growth, 0.53 points are labelled “capital” and 0.97 points are the Solow residual. ✓

Consistency check. Rewrite the production function in Hicks-neutral form, $Y = \tilde{A}K^\alpha L^{1-\alpha}$ with $\tilde{A} \equiv A^{1-\alpha}$. Then $g_{\tilde{A}} = (1 - \alpha)g$, which is exactly the residual. The two ways of writing technology give the same accounting, as they must. ✓

How this relates to “no technological progress, no growth”. It looks like a contradiction: Proposition 4.3 says technology is the *only* source of long-run growth, yet the accounting gives it only 65% of the credit. Both statements are true, because they answer different questions.

- Growth accounting is a **proximate** decomposition. It asks: holding the other inputs fixed, how much of the observed change in output does each observed change in an input mechanically explain? It treats g_K as a fact about the world, with no view on where it came from.
- The Solow model gives the **ultimate** cause. Here capital grows *only because* technology grows: $g_K = g$ is not an independent parameter, it is an equilibrium response. Set $g = 0$ and the whole thing collapses — $g_K = 0$ and $g_Y = 0$. Every one of the 0.53 points labelled “capital” would disappear.

Formally, the total derivative of steady-state growth with respect to g is

$$\frac{dg_Y}{dg} = \underbrace{\alpha \frac{dg_K}{dg}}_{\alpha \cdot 1} + \underbrace{(1 - \alpha)}_{\text{direct}} = 1,$$

against a partial derivative, holding g_K fixed, of only $1 - \alpha$. Growth accounting reports the partial; the model reports the total.

Reading. This is the standard caution about the Solow residual, and it runs in both directions. Growth accounting *understates* the role of anything that induces capital accumulation — technology here, but equally an institutional reform or a fall in the cost of investment goods. It *overstates* the role of capital in exactly the same amount. A decomposition that gives capital 35% of Soviet growth (Example 5.1) is not a claim that capital was 35% of the reason; it is a claim about the arithmetic of a Taylor expansion. One can convert a proximate share into an ultimate one by dividing the residual by $1 - \alpha$: here $(1 - \alpha)g / (1 - \alpha) = g$, technology gets 100%, which is Proposition 4.3 recovered.

Problem 5.6 Interest Rates

Question

Suppose we observe that Usuria (a closed economy) grows at approximately 6% a year, and we are trying to understand why. We know the labor force has been constant.

- **Conjecture 1:** The economy has been at a steady-state-with-technological-progress. There has been capital accumulation just to maintain $K/(AL)$ constant but the cause of growth has been TFP growth.

- **Conjecture 2:** The economy started from a very low level of capital stock (below steady state) and has been growing because it is converging to the steady state, but TFP has been constant.

Ideally, if we wanted to distinguish between Conjecture 1 and Conjecture 2 we could do a growth-accounting exercise. Unfortunately, Usuria does not collect reliable statistics on capital accumulation that would enable us to do this. We do, however, have data on interest rates in Usuria. How would one use these data to distinguish between Conjecture 1 and Conjecture 2? Be as mathematically precise as possible.

The idea in one line. The interest rate *is* a measurement of the capital stock. We cannot see K , but r is a monotone function of K/Y , so the behaviour of r tells us what K is doing.

Step 1: the link. With $Y = K^\alpha (AL)^{1-\alpha}$ and competitive factor markets,

$$r_t = F_K(K_t, A_t L_t) - \delta = \alpha K_t^{\alpha-1} (A_t L_t)^{1-\alpha} - \delta = \alpha \frac{Y_t}{K_t} - \delta, \quad \text{i.e.} \quad \frac{K_t}{Y_t} = \frac{\alpha}{r_t + \delta}. \quad (5.6.1)$$

This holds on *any* path, in or out of steady state, and requires no data on K itself.

Step 2: what each conjecture implies. Take logs of (5.6.1) and differentiate:

$$\frac{d \ln(r_t + \delta)}{dt} = g_Y - g_K. \quad (5.6.2)$$

- **Conjecture 1.** On a balanced growth path $K/(AL)$ is constant and L is constant, so $g_K = g_A = g_Y = 6\%$. Then (5.6.2) gives

$$\frac{d \ln(r + \delta)}{dt} = 0 \quad \implies \quad \boxed{r_t \text{ is constant}}.$$

- **Conjecture 2.** A is constant and L is constant, so all of the growth in Y must come from K . Growth accounting with a zero residual gives

$$g_Y = \alpha g_K \quad \implies \quad g_K = \frac{g_Y}{\alpha} = \frac{0.06}{0.35} = 17.1\% \text{ per year},$$

and (5.6.2) becomes

$$\frac{d \ln(r + \delta)}{dt} = g_Y \left(1 - \frac{1}{\alpha}\right) = 0.06 \times (1 - 2.857) = \boxed{-11.1\% \text{ per year}}.$$

Step 3: the test. Regress $\ln(r_t + \delta)$ on time.

$$\text{slope} = 0 \Rightarrow \text{Conjecture 1}; \quad \text{slope} = g_Y \left(1 - \frac{1}{\alpha}\right) < 0 \Rightarrow \text{Conjecture 2}.$$

The two predictions are not merely different in sign, they are far apart in magnitude: under Conjecture 2 the gross return $r + \delta$ falls by a factor $e^{-1.11} = 0.33$ over a single decade. Ten years of interest-rate data settle the question, and the only auxiliary parameters needed are α (readable off the labour share, which Usuria does report as part of its GDP accounts) and δ , which enters only through the level shift $r + \delta$.

A sharper version using levels. If we are also willing to use the investment rate s , the two conjectures predict very different interest rates *today*. From $g_K = sY/K - \delta$ and (5.6.1),

$$r = \alpha \frac{g_K + \delta}{s} - \delta.$$

Taking the §5.2 values $s = 0.2$, $\delta = 0.04$, $\alpha = 0.35$:

	g_K	implied r
Conjecture 1 (balanced growth, $g_K = g_Y$)	6.0%	13.5%
Conjecture 2 (transition, $g_K = g_Y/\alpha$)	17.1%	33.0%

✓ A 33% real interest rate is not something one would fail to notice. The corresponding K/Y ratios are 2.6 and 0.95: under Conjecture 2 the economy would have less than one year of GDP in capital, against a US figure of about three.

Reading. This exercise is a small lesson in identification. Two stories generate the same observable — 6% growth — and are distinguished by a *different* observable that the theory ties to the unobserved variable. It is also a reality check on Conjecture 2 generally: a 17% annual capital growth rate sustained for years is enormous, and the implied interest rate is far above anything ever measured. That is essentially the argument Lucas (1990) makes about why capital does not flow to poor countries, and it is another way of stating why §5.3 rejects the capital-accumulation hypothesis.

National Accounts, Factor Shares and the Golden Rule

Problem 5.7 GDP Accounting, Interest Rates and Growth Accounting

Question

Proletaria is a mythical country in Central Asia, where the currency is the ruble. The exchange rate is 10 rubles per euro. Suppose the production function in Proletaria is given by $Y = K^\alpha L^{1-\alpha}$.

(a) Find expressions for the output-to-capital ratio Y/K and the marginal product of capital F_K as functions of K and L .

During 2014, the following events took place in Proletaria.

- Kapitas, the main manufacturer in the country, imported an industrial welder made in Germany, for which it paid 1,000 euro.
- By the end of the year, the industrial welder was no longer new. Its estimated value in the resale market was 960 euro.
- 100 workers worked for Kapitas the entire year making screws and nails, using the new welder. Each of them received wages for 350 rubles.
- Each of the Kapitas workers paid 100 rubles in income taxes.
- The total output of Kapitas consisted of 1 million screws and 1 million nails. All of it was exported to Austria for a total of 4,000 euro.
- The government of Proletaria employed one of the 100 workers as Chief of the Secret Police (in addition to his factory job) to maintain law and order, and paid her a salary of 10,000 rubles.
- The workers ate beef imported from France, which cost a total of 2,500 euro.

(b) Construct GDP accounts (in rubles) for Proletaria by production, income and expenditure.

(c) What were the income shares of labor and capital?

(d) Suppose that we know that the capital stock in Proletaria is 100,000 rubles and that the depreciation rate of the industrial welder is typical for this country's capital stock. What interest rate should we expect to observe?

(e) During 2015, additional investment in Proletaria has exceeded depreciation so that the capital stock now stands at 120,000 rubles. The labor force also grew thanks to immigration, and now consists of 105 workers instead of 100. GDP during 2015 was 55,000 rubles (prices were constant). How much of the growth in GDP between 2014 and 2015 can be attributed to growth in Total Factor Productivity?

Solution:

(a) Divide the production function by K and differentiate it:

$$\frac{Y}{K} = K^{\alpha-1} L^{1-\alpha} = \left(\frac{L}{K}\right)^{1-\alpha}, \quad F_K = \alpha K^{\alpha-1} L^{1-\alpha} = \alpha \left(\frac{L}{K}\right)^{1-\alpha} = \alpha \frac{Y}{K}.$$

The second identity, $F_K = \alpha Y/K$, is the workhorse of parts (d) and (e): with Cobb-Douglas, the marginal product of capital is the capital share times the output-capital ratio, and both are observable.

- (b) Convert everything at 10 rubles per euro: the welder costs 10,000 rubles and is worth 9,600 at year-end; the exports are 40,000; the beef is 25,000.

Production (value added).

Producer	Gross output	Value added
Kapitas (screws and nails)	40,000	40,000
Government (police services, valued at cost)	10,000	10,000
GDP		50,000

Kapitas uses no intermediate inputs: the welder is *capital*, not an intermediate good, so it is not netted out of value added. The beef and the welder are produced abroad and contribute nothing to Proletarian production.

Income.

Wages, Kapitas (100×350)	35,000
Wages, government (the police chief's salary)	10,000
Capital income / operating surplus, Kapitas ($40,000 - 35,000$)	5,000
GDP	50,000

Expenditure.

C (beef)	25,000
I (welder)	10,000
G (police services)	10,000
X (screws and nails)	40,000
$-M$ (welder 10,000 + beef 25,000)	-35,000
GDP	50,000

✓ The three methods agree at 50,000 rubles, which is the check the exercise is built around.

Four traps in this list, and each is deliberate.

- **Income taxes (10,000 rubles) are not in GDP anywhere.** They are a transfer from households to the government; the production they finance is already counted at the salaries paid. Adding them would double-count.
- **The police chief's services are counted at cost, 10,000 rubles, because they are not sold in a market.** This is the convention of §1.3 and it is why the largest single item of Proletarian value added is a service with no price.
- **The police chief holds two jobs.** She is one of the 100 factory workers, so the labour force is 100, not 101. Her factory wage (350) is inside the 35,000 and her government salary (10,000) is on top of it.
- **Depreciation (400 rubles) does not appear.** These are *gross* accounts; the 400 would be subtracted to get NDP, and it is used in part (d) to pin δ .

- (c) Read the shares off the income account:

$$\text{labour share} = \frac{35,000 + 10,000}{50,000} = \boxed{0.90}, \quad \text{capital share} = \alpha = \frac{5,000}{50,000} = \boxed{0.10}.$$

The shares sum to one, as the income method requires. ✓

A capital share of 0.10 is very low — the empirical benchmark is 0.35 — and the reason is visible in the accounts: a fifth of Proletarian GDP is a government service valued at the wage bill, which is by construction 100% labour income. Toy economies with a large imputed public sector always look labour-intensive.

- (d) Depreciation of the welder: it lost $10,000 - 9,600 = 400$ rubles of value on a stock of 10,000, so

$$\delta = \frac{400}{10,000} = 0.04.$$

Then, using $F_K = \alpha Y/K$ from part (a),

$$r = F_K - \delta = \alpha \frac{Y}{K} - \delta = 0.10 \times \frac{50,000}{100,000} - 0.04 = 0.05 - 0.04 = \boxed{1\%}.$$

✓

The number is low, and part (c) explains why: with a capital share of only 0.10, capital earns 5,000 a year on a stock of 100,000, a gross return of 5%, of which four points are eaten by depreciation. Note also the internal consistency: $K/Y = 2$ and $\alpha/(r + \delta) = 0.10/0.05 = 2$. ✓

- (e) Growth accounting, equation (5.4.2), with the shares from part (c):

$$g_Y = \frac{55,000}{50,000} - 1 = 10\%, \quad g_K = \frac{120,000}{100,000} - 1 = 20\%, \quad g_L = \frac{105}{100} - 1 = 5\%,$$

$$\text{TFP} = g_Y - \alpha g_K - (1 - \alpha)g_L = 0.10 - 0.10 \times 0.20 - 0.90 \times 0.05,$$

$$\text{TFP} = 0.10 - 0.02 - 0.045 = \boxed{3.5\%}.$$

✓ So 3.5 of the 10 percentage points of growth — **35% of it** — is Total Factor Productivity.

Reading. Note how much the answer depends on part (c). Capital grew twice as fast as output, which sounds like the dominant force, but it is weighted by a share of only 0.10 and contributes just 2 points. Labour grew half as fast but carries a weight of 0.90 and contributes more than twice as much. This is the whole content of “factor shares are the right weights”: the fast-growing input is not necessarily the important one.

Problem 5.8 National Accounts and The Golden Rule

Question

In 2018, the following events happened in Aurum, which is a closed economy that uses the Denarius (plural: Denarii) as its currency.

- Grano, Inc. hired Cornelia to plant and harvest 500 tons of wheat on a large plot of land using a tractor. Both the land and the tractor are owned by Grano, Inc. It paid Cornelia 200 Denarii for her work on the harvest. It sold the wheat to Panem, Inc. for 1,000 Denarii.
- The tractor Cornelia used was new at the beginning of the year, worth 2,000 Denarii, and was no longer new at the end of the year: by then it was worth 1,800 Denarii.
- Panem, Inc. hired Gaius to grind the wheat in its mill and produce a million loaves of bread. It paid Gaius 1,400 Denarii for his work. It sold the bread for 4,000 Denarii to the hungry citizens, who eat it.
- Fabrica, Inc. hired Livia to build a combine harvester. It paid Livia 400 Denarii for her work. It sold the combine harvester to Grano, Inc. for 1,000 Denarii.

(a) Construct GDP accounts (in Denarii) for Aurum by production, income and expenditure.

(b) What were the income shares of labor and capital?

(c) What was the savings rate?

(d) Assume the following: the savings rate is constant; the production function has the form $Y = AK^\alpha L^{1-\alpha}$ and there is no technological progress; the depreciation rate for the tractor is typical for the capital stock as a whole; the population is constant. In the long run, would the level of consumption be higher if the economy slightly increased its saving rate?

Solution:

(a) **Production (value added).**

Firm	Gross output	Intermediates	Value added
Grano (wheat)	1,000	0	1,000
Panem (bread)	4,000	1,000	3,000
Fabrica (combine harvester)	1,000	0	1,000
GDP			5,000

The wheat is an *intermediate* good: it is bought by Panem and physically transformed within the year, so it is netted out of Panem's value added. The combine harvester is *capital*: it is bought by Grano but survives the year, so it is not netted out of anything and stays in GDP as investment.

Income.

	Wages		Capital income	Total
Grano	200		$1,000 - 200 = 800$	1,000
Panem	1,400	$4,000 - 1,000 - 1,400 = 1,600$		3,000
Fabrica	400		$1,000 - 400 = 600$	1,000
Total	2,000		3,000	5,000

Capital income here is gross operating surplus: it is the return to the tractor, the land and the mill, and it includes the 200 of depreciation.

Expenditure.

C (bread, eaten by citizens)	4,000
I (combine harvester)	1,000
$G = 0$, $X = M = 0$ (closed economy, no government)	0
GDP	5,000

✓ Three methods, 5,000 Denarii each.

Where this goes wrong

The wheat and the combine harvester cost the same 1,000 Denarii and are treated in opposite ways. The test is not the price and not the buyer — both are bought by firms — but whether the good *survives the period*. Wheat is consumed in production this year (intermediate, netted out); a harvester is used for years (capital, counted as investment). Netting out the harvester would give $GDP = 4,000$; counting the wheat would give 6,000. Only 5,000 makes the three methods agree.

(b)

$$\text{labour share} = \frac{2,000}{5,000} = \boxed{0.4}, \quad \text{capital share} = \alpha = \frac{3,000}{5,000} = \boxed{0.6}.$$

(c) In a closed economy with no government, $S = Y - C$, and in equilibrium $S = I$:

$$s = \frac{Y - C}{Y} = \frac{5,000 - 4,000}{5,000} = \frac{I}{Y} = \frac{1,000}{5,000} = \boxed{0.2}.$$

(d) **Yes.** The economy is below the Golden Rule, so raising s raises long-run consumption.

Step 1: the parameters. From (b), $\alpha = 0.6$. From (c), $s = 0.2$. The tractor lost 200 of a stock of 2,000, so

$$\delta = \frac{200}{2,000} = 0.1,$$

and $n = g = 0$ by assumption.

Step 2: where the economy is. In steady state $sY = \delta K$, so

$$\left. \frac{K}{Y} \right|_{ss} = \frac{s}{\delta} = \frac{0.2}{0.1} = 2, \quad \text{MPK} = \alpha \frac{Y}{K} = \frac{0.6}{2} = 0.30.$$

Step 3: where the Golden Rule is. The Golden Rule maximises steady-state consumption $c^* = f(k^*) - \delta k^*$, giving $f'(k_{gr}) = \delta$, i.e.

$$\text{MPK} = \delta \iff \alpha \frac{Y}{K} = \delta \iff \left. \frac{K}{Y} \right|_{gr} = \frac{\alpha}{\delta} = \frac{0.6}{0.1} = 6,$$

and since $K/Y = s/\delta$ in steady state, the Golden Rule saving rate is

$$s_{gr} = \alpha = 0.6. \quad (5.8.1)$$

Step 4: compare.

$$\text{MPK} = 0.30 > \delta = 0.10, \quad \text{equivalently} \quad s = 0.2 < s_{gr} = \alpha = 0.6.$$

The economy is **under**-accumulating capital, so

yes — a slightly higher saving rate raises long-run consumption.

Verification. With $y^* = A^{1/(1-\alpha)}(s/\delta)^{\alpha/(1-\alpha)}$,

$$c^*(s) = (1-s) A^{\frac{1}{1-\alpha}} \left(\frac{s}{\delta} \right)^{\frac{\alpha}{1-\alpha}} \propto (1-s) s^{1.5},$$

whose derivative $\frac{d}{ds} [(1-s)s^{1.5}] = s^{0.5}(1.5 - 2.5s)$ is positive for $s < 0.6$ and zero at $s = 0.6 = \alpha$. ✓ A grid search over $s \in (0, 1)$ in `ch05_numerico.py` finds the maximum at $s = \alpha$ to four digits, and the finite-difference derivative at $s = 0.2$ is strictly positive. ✓ Going all the way to the Golden Rule would raise long-run consumption by a factor of 2.6.

Reading, and two cautions.

- Result (5.8.1) — *the Golden Rule saving rate equals the capital share* — is worth memorising, and it is the cleanest way to test dynamic efficiency in data: an economy saves too much only if its investment rate exceeds its capital share. Actual investment rates are 20–30% and actual capital shares 30–40%, which is why no real economy has ever been convincingly shown to be dynamically inefficient. Aurum’s $\alpha = 0.6$ is a toy artefact of a three-firm economy, and it makes the gap look far bigger than it is anywhere real.
- “Higher long-run consumption” is not “better”. Getting to the higher steady state requires consuming *less* for many years first, and nothing in the Solow model says the trade is worth taking — the model has no preferences to evaluate it with. That is exactly the gap Chapter 9 fills: with an optimising household the saving rate is chosen, and Problem 9.11 shows the resulting steady state lies strictly *below* the Golden Rule, on purpose. The Golden Rule is a benchmark, not a target.

Mismeasurement as a Source of Measured TFP

Problem 5.4 Measurement of the Capital Stock

Question

The capital stock is not easy to measure. A common way to try to do this is to guess a value for K_0 at some date in the past, measure investment every year thereafter, and use the equation

$$K_{t+1} = (1 - \delta)K_t + I_t \quad (5.5.1)$$

to compute the capital stock for every year thereafter. Consider an economy that is in a steady state without technological progress (and has been there for an infinitely long time). Every year the level of investment is $I_t = 0.2$ and the depreciation rate is $\delta = 0.1$.

- What is the capital stock?
- The statistical office only started measuring investment in year 0. By then the economy was already in steady state. They guessed (incorrectly) that the capital stock in year 0 was $K_0 = 1$. Use equation (5.5.1) to compute how the estimate of the capital stock changed over time after year 0 and plot your answer. Compute the ratio $K^{\text{EST}}/K^{\text{TRUE}}$ for the years 5, 10 and 50.
- Suppose now that the statistics office started collecting statistics a long time ago (an infinite time ago) so we don't have the problem of making the wrong initial guess. However, instead of using the correct value of δ , the statistical office incorrectly believes that $\delta = 0.05$. What is their estimate of the capital stock? Why does it differ from the truth?
- The production function in this economy is $Y = AK^\alpha L^{1-\alpha}$ and there are accurate data on GDP and on total hours of labor supply. Using the inaccurate estimate of K from part (c), plus accurate data on total labor and GDP, an economist is trying to measure the economy's productivity. Find an expression for $A^{\text{EST}}/A^{\text{TRUE}}$ as a function of $K^{\text{EST}}/K^{\text{TRUE}}$. Given the numbers from part (c) and using $\alpha = 0.35$, how far would the economist's estimate be from the truth?

Solution:

- In a steady state K is constant, so (5.5.1) gives $K = (1 - \delta)K + I$, i.e. $\delta K = I$:

$$K^{\text{TRUE}} = \frac{I}{\delta} = \frac{0.2}{0.1} = 2.$$

- Iterating (5.5.1) from a wrong initial value is a linear difference equation with the same fixed point. Write $e_t \equiv K_t^{\text{EST}} - K^{\text{TRUE}}$; subtracting the two recursions,

$$e_{t+1} = (1 - \delta)e_t \quad \implies \quad e_t = (1 - \delta)^t e_0 = -(0.9)^t,$$

so the estimate has the closed form

$$\boxed{K_t^{\text{EST}} = 2 - (0.9)^t}, \quad \frac{K_t^{\text{EST}}}{K^{\text{TRUE}}} = 1 - \frac{1}{2}(0.9)^t.$$

✓ (verified against the raw iteration in `ch05_numerico.py`).

t	K_t^{EST}	$K_t^{\text{EST}}/K^{\text{TRUE}}$
5	1.4095	0.7048
10	1.6513	0.8257
50	1.9948	0.9974

The error decays geometrically at rate $1 - \delta = 0.9$, with a half-life of $\ln 2 / \ln(1/0.9) = 6.6$ years. After five decades it is gone.

- (c) With an infinite history and no initial-condition problem, repeated substitution of (5.5.1) gives the perpetual-inventory formula

$$K^{\text{EST}} = \sum_{j=0}^{\infty} (1 - \hat{\delta})^j I = \frac{I}{\hat{\delta}},$$

so

$$\boxed{K^{\text{EST}} = \frac{0.2}{0.05} = 4}, \quad \frac{K^{\text{EST}}}{K^{\text{TRUE}}} = \frac{\delta}{\hat{\delta}} = \frac{0.10}{0.05} = \boxed{2}.$$

The estimate is **double** the truth. The reason is mechanical: the office believes each vintage of investment survives twice as long as it really does, so it keeps summing old investment that has in fact already worn out. ✓

- (d) Invert the production function for A , using the same (accurate) Y and L in both:

$$A = \frac{Y}{K^\alpha L^{1-\alpha}} \quad \Rightarrow \quad \frac{A^{\text{EST}}}{A^{\text{TRUE}}} = \frac{Y / [(K^{\text{EST}})^\alpha L^{1-\alpha}]}{Y / [(K^{\text{TRUE}})^\alpha L^{1-\alpha}]} = \left(\frac{K^{\text{EST}}}{K^{\text{TRUE}}} \right)^{-\alpha}.$$

With the ratio 2 from part (c) and $\alpha = 0.35$,

$$\frac{A^{\text{EST}}}{A^{\text{TRUE}}} = 2^{-0.35} = \boxed{0.7846},$$

so measured productivity is 21.5% **below** the truth. ✓

Reading. Three things, in order of importance.

1. **The two errors are not equally serious, and it is the harmless-looking one that is fatal.** A wrong initial guess is self-correcting: whatever you assume, the recursion forgets it geometrically, and this is precisely why statistical agencies can get away with an arbitrary benchmark. A wrong depreciation rate is a *permanent, proportional* bias: it never washes out, no matter how long the series. Figure 3 draws the contrast.

2. **The sign is counterintuitive.** Overstating capital *understates* TFP, because the residual is what output is left over after the inputs have been credited. Credit the inputs with too much and there is too little left for technology.
3. **The damage is muted by α .** The elasticity is $-\alpha = -0.35$, so a 100% error in K produces only a 21.5% error in A . Since δ is the hardest parameter in the whole exercise — the BEA's own figures range from 0.02 for buildings to 0.3 for computers (§5.2) — this is reassuring, and it is one reason cross-country TFP comparisons survive the crudeness of the capital data. It does not rescue them from the errors in *labour* input, which Problems 5.5 and 5.9 show come through with an elasticity of one.

5.4 — two ways of getting the capital stock wrong ($I = 0.2$, $\delta = 0.1$)

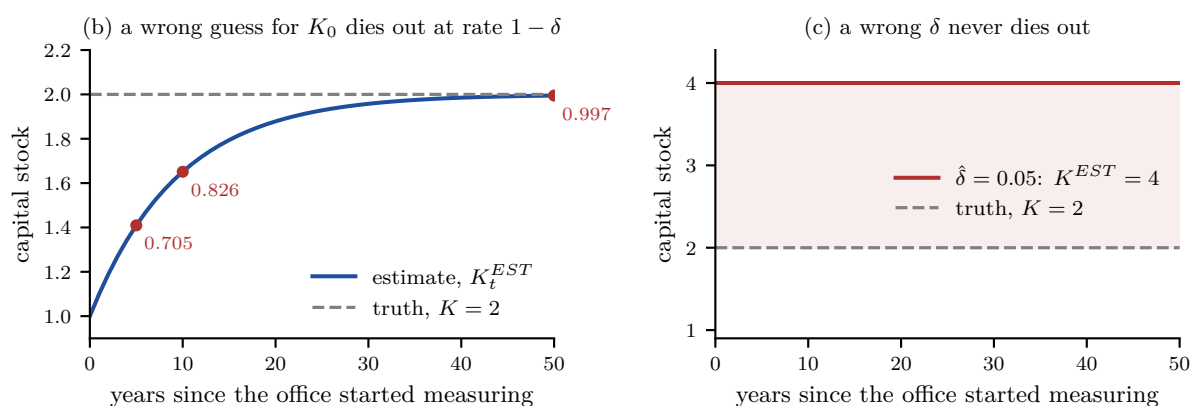


Figure 3: Problem 5.4. A wrong initial guess (left) is forgotten at rate $1 - \delta$; a wrong depreciation rate (right) never is.

Problem 5.5 Sources of TFP

Question

Suppose that the true production function in Gotham is

$$Y = AK^\alpha L^{1-\alpha} \quad (5.5.3)$$

Unfortunately, crime is a huge problem in Gotham, so that for each worker doing actual work firms need to hire ϕ security guards just to protect their products from being stolen. The security guards will of course describe their activity as work even though they are not actually producing anything. Use the notation N to refer to the total labor force (including workers and guards) and denote the number of actual production workers by L .

- (a) Find an expression for total output as a function of A , K , N , ϕ and α .
- (b) Write down the problem of a firm that has to choose capital and labor to maximize profits. Notice that the firm will have to pay a wage to the security guards even though they will not produce anything.
- (c) If the representative firm hires all the workers and rents all the capital, what will be the wage and the rental rate of capital? Express it as a function of A , K , N , ϕ and α .
- (d) Suppose an economist studying Gotham is trying to estimate A using equation (5.5.3). The economist has accurate data on K , N and Y . However, the economist doesn't really know whether workers are involved in production or in security services: in national statistics they all look employed. Therefore the economist will plug in the value of N instead of the value of L into the estimate of A . What will the economist's estimate of A be? How does it compare to the true value of A ?
- (e) How does this relate to the findings that link GDP levels to social and political institutions?
- (f) Suppose that, in an economy that didn't have the crime problem of Gotham, the government attempted to create jobs by mandating that firms hire ϕ assistants for every production worker. The job of assistants is to look at production workers all day long and not do anything. Using the analysis above, what would be the effects of such a policy?

Solution:

- (a) Each production worker requires ϕ guards, so a firm employing L producers must hire $L + \phi L$ people in total:

$$N = (1 + \phi)L \quad \implies \quad L = \frac{N}{1 + \phi}.$$

Substituting into (5.5.3),

$$Y = AK^\alpha \left(\frac{N}{1+\phi} \right)^{1-\alpha} = \underbrace{A(1+\phi)^{-(1-\alpha)}}_{\tilde{A}} K^\alpha N^{1-\alpha} \quad (5.5.a)$$

The economy behaves *exactly* like one with the same total employment N and a lower Hicks-neutral productivity $\tilde{A} = A(1+\phi)^{-(1-\alpha)}$. That observational equivalence is the whole exercise.

- (b) The firm hires N people at the market wage w (it must pay guards too) and rents K at r^K . Using (5.5.a),

$$\max_{K, N} A(1+\phi)^{-(1-\alpha)} K^\alpha N^{1-\alpha} - wN - r^K K.$$

It is important that the firm's choice variable is N , not L : the technology ϕ is imposed, so the firm cannot economise on guards, and every person on the payroll costs w .

- (c) First-order conditions:

$$w = (1-\alpha) A(1+\phi)^{-(1-\alpha)} \left(\frac{K}{N} \right)^\alpha, \quad r^K = \alpha A(1+\phi)^{-(1-\alpha)} \left(\frac{N}{K} \right)^{1-\alpha}.$$

Both are the usual Cobb-Douglas factor prices with A replaced by $\tilde{A}$. Two remarks:

- **Guards and producers earn the same wage w** , even though the guards produce nothing. In a competitive labour market a firm cannot pay two prices for the same labour, and it will not hire a guard unless the guard is worth w to it — which he is, because without him the goods are stolen. The guard's marginal product to the *firm* is w ; his marginal product to *society* is zero.
- Factor shares are unaffected: $wN/Y = 1-\alpha$ and $r^K K/Y = \alpha$ exactly as before. ✓ Crime is therefore invisible in the income accounts, which is why part (d) works.

- (d) The economist computes

$$\hat{A} = \frac{Y}{K^\alpha N^{1-\alpha}} \stackrel{(5.5.a)}{=} A(1+\phi)^{-(1-\alpha)} \implies \boxed{\frac{\hat{A}}{A} = (1+\phi)^{-(1-\alpha)} < 1}.$$

Measured TFP is *below* true TFP. With $\alpha = 0.35$ and $\phi = 1$ (one guard per worker), $\hat{A}/A = 2^{-0.65} = 0.637$: Gotham looks 36% less productive than it is. ✓

It is worth pushing this to the long run, because that is where the number becomes interpretable. In a Solow steady state with $Y = \tilde{A}K^\alpha N^{1-\alpha}$ and $n = g = 0$,

$$\left. \frac{Y}{N} \right|_{ss} = \tilde{A}^{\frac{1}{1-\alpha}} \left(\frac{s}{\delta} \right)^{\frac{\alpha}{1-\alpha}}, \quad \tilde{A}^{\frac{1}{1-\alpha}} = A^{\frac{1}{1-\alpha}} \frac{1}{1+\phi},$$

so long-run GDP per person, *and the wage with it*, fall by exactly

$$\frac{1}{1 + \phi},$$

which is one half when $\phi = 1$. ✓ The exponents cancel and leave something transparent: in the long run the economy loses output in exact proportion to the fraction of its labour force that is not producing anything. The $(1 - \alpha)$ in the TFP formula is an artefact of holding K fixed; once capital adjusts, the loss is one-for-one.

- (e) **This is a mechanism, not just an analogy.** The cross-country evidence (§5.5) finds that GDP per capita correlates with rule of law, property-rights protection, control of corruption and political stability. The natural objection is: institutions are not a production function, so how can they show up in A ? Gotham is the answer. Nothing about Gotham’s technology is worse — A is the same, α is the same, the blueprints are the same. What differs is that weak protection of property forces the economy to spend a fraction $\phi/(1 + \phi)$ of its labour on *guarding* output rather than producing it, and the national accounts have no way to see the difference. The measured Solow residual falls, and an economist with no other information calls it “low TFP”.

Two extensions, both in the chapter:

- The same argument covers any diversion of resources into activities that are privately valuable and socially unproductive: litigation, tax avoidance, bribery, lobbying, duplicated regulatory compliance. This is the theory of rent-seeking, and it predicts that measured TFP is a *lower bound* on technological competence.
 - Hsieh and Klenow (2009) is the same idea applied to *allocation* rather than diversion: when capital and labour sit in the wrong firms, aggregate output falls for given aggregate inputs, and the accounts record it as TFP. Their estimate is that eliminating Chinese and Indian misallocation down to US levels would raise measured TFP by 30–50% and 40–60%.
- (f) **A make-work mandate is arithmetically identical to crime, and it does not create jobs.** Replace “guards” with “assistants”; every equation above goes through unchanged. Therefore:
- **Employment is unchanged.** N is the labour force, and it was already fully employed. The mandate does not create a job, it relabels $\phi/(1 + \phi)$ of the existing ones. (Nothing in this model can generate unemployment; that machinery arrives in Chapter 7. But that is the point — the policy’s stated benefit does not exist even in principle here.)
 - **Output falls**, immediately by $(1 + \phi)^{-(1-\alpha)}$ and in the long run by $1/(1 + \phi)$.
 - **Wages fall in the same proportion**, since $w = (1 - \alpha)\tilde{A}(K/N)^\alpha$ and $\tilde{A}$ has

fallen. With $\phi = 1$ the long-run wage is *halved*. ✓ The intended beneficiaries pay for the policy in full.

- **Measured productivity falls** by $(1 + \phi)^{-(1-\alpha)}$, so the statistics would report a technological regression that never happened.

There is one asymmetry with Gotham worth stating, and it is the policy conclusion. Gotham's guards are a *second-best response to a real problem*: given that theft exists, hiring them is efficient, and the right intervention is to attack the crime, not the guards. Assistants respond to no problem at all, so the policy is pure deadweight loss. The observable consequences are identical, which is precisely why one cannot read policy recommendations off a Solow residual.

Problem 5.9 Disease and TFP

Question

The production function in Influenzistan is

$$Y = K^\alpha (AH)^{1-\alpha} \quad (5.5.4)$$

where H is the total number of hours of work.

- Solve equation (5.5.4) for A , so that you end up with an expression for A in terms of all the other variables and parameters.
- Suppose an economist wants to measure A . The economist has official data from the Influenzistan Statistics Office on: total hours worked by all workers; the total level of investment, which has been constant for many many years; an estimate of the average rate of depreciation of the capital stock; and GDP and its subcomponents, measured by the income method. How can he use this data to get an estimate of A ? Don't be vague: provide exact formulas that describe each step precisely.
- Suppose now that the economist has the same data as above except he doesn't know total hours worked. Instead, the Statistics Office only keeps records on the total number of employed workers (call this number L), but not on how many hours each of them works. In order to get an estimate of A , the economist assumes that the average worker works N hours per year, which is the number of hours worked by the average worker in neighboring Healthistan. Write down an expression for the economist's estimate of A (call this number $\hat{A}$).
- Unfortunately, the residents of Influenzistan keep getting sick, so they have to spend a significant portion of their time recovering at home instead of working. As a result, they each work xN hours instead of N hours, where $x < 1$. Derive an expression for $\hat{A}/A$, i.e. for the ratio of the economist's estimate of A to its true value. Interpret your finding.

Solution:

(a) Isolate the A term and raise to the power $1/(1 - \alpha)$:

$$Y = K^\alpha A^{1-\alpha} H^{1-\alpha} \implies A^{1-\alpha} = \frac{Y}{K^\alpha H^{1-\alpha}} \implies A = \frac{1}{H} \left(\frac{Y}{K^\alpha} \right)^{\frac{1}{1-\alpha}} \quad (5.9.a)$$

or, written out, $A = Y^{\frac{1}{1-\alpha}} K^{-\frac{\alpha}{1-\alpha}} / H$.

Note the exponent on H : it is exactly -1 . Because A is *labour-augmenting*, it multiplies H one-for-one, so any proportional error in H becomes the same proportional error in A . In Problem 5.4 the corresponding elasticity, of measured productivity with respect to the capital estimate, was only $-\alpha = -0.35$; here it is -1 . That gap is the point of the whole problem.

(b) Four steps, each tied to one of the four data items.

Step 1. Get α from the income accounts. The income-method subcomponents split GDP into labour and capital income. Under competition the capital share equals the output elasticity of capital, so

$$\alpha = \frac{\text{capital income}}{Y} = \frac{\text{operating surplus}}{\text{operating surplus} + \text{compensation of employees}}.$$

This is why the exercise insists on the *income* method: no other decomposition of GDP delivers α .

Step 2. Get K from investment and depreciation. Investment has been constant at I “for many many years”, so the capital stock has converged to its steady state, where $\delta K = I$ (this is exactly Problem 5.4(a)):

$$\hat{K} = \frac{I}{\delta}.$$

Constancy of I is what makes this legitimate — it removes the need for a whole investment history and for an initial condition.

Step 3. Read Y and H directly off the accounts and the hours survey.

Step 4. Substitute into (5.9.a):

$$\hat{A} = \frac{1}{H} \left(\frac{Y}{(I/\hat{\delta})^\alpha} \right)^{\frac{1}{1-\alpha}}$$

✓ `problema_5_9()` builds an artificial economy from a known A , runs these four steps, and recovers A exactly.

(c) Replace the unobserved H by the assumed $\hat{H} = LN$:

$$\hat{A} = \frac{1}{LN} \left(\frac{Y}{(I/\hat{\delta})^\alpha} \right)^{\frac{1}{1-\alpha}} = \frac{Y^{\frac{1}{1-\alpha}} K^{-\frac{\alpha}{1-\alpha}}}{LN}$$

- (d) The true hours are $H = LxN$. Since Y , K and α are measured correctly, everything in the numerator of (5.9.a) is common to $\hat{A}$ and A and cancels:

$$\frac{\hat{A}}{A} = \frac{Y^{\frac{1}{1-\alpha}} K^{-\frac{\alpha}{1-\alpha}} / (LN)}{Y^{\frac{1}{1-\alpha}} K^{-\frac{\alpha}{1-\alpha}} / (LxN)} = \frac{LxN}{LN} = \boxed{x}$$

✓

Interpretation — and this is the sharpest version of the chapter's warning.

1. **The bias is one-for-one and α does not appear.** If sickness costs 20% of working hours, measured TFP is 20% too low — not $0.8^{1-\alpha}$, not 0.8^α , but exactly 0.8. There is no dilution and no offsetting term. Compare Problem 5.4, where doubling the capital estimate cost only 21.5% of measured A : errors in *labour* input are far more damaging to a residual than errors in capital, because labour carries the larger share and, in the labour-augmenting formulation, A absorbs the error directly.
2. **The disease is real and the productivity gap is not.** Influenzistan really is poorer: fewer hours are worked, so Y really is lower. What is spurious is the *attribution*. An economist comparing Influenzistan with Healthistan would report a technology gap of $1 - x$, when the two countries have identical technology and differ only in how much labour they supply. The measurement error and the underlying problem point in the same direction, which is what makes the mistake hard to catch.
3. **Consistency with Problem 5.5.** ✓ The two exercises are the same trap in different notation. In Gotham, labour input was over-counted by a factor $m = 1 + \phi$ and the Hicks-neutral $\tilde{A}$ was biased by $m^{-(1-\alpha)}$. Here labour is over-counted by $m = 1/x$. Converting the labour-augmenting A into its Hicks-neutral equivalent $\tilde{A} = A^{1-\alpha}$,

$$\frac{\hat{\tilde{A}}}{\tilde{A}} = \left(\frac{\hat{A}}{A} \right)^{1-\alpha} = x^{1-\alpha} = m^{-(1-\alpha)},$$

which is Gotham's formula exactly. ✓ This identity is asserted in `ch05_numeric.py` and would abort the script if it failed. The apparent disagreement between “bias = x ” and “bias = $m^{-(1-\alpha)}$ ” is entirely about which A one is talking about.

4. **The practical moral.** This is why the development-accounting literature works so hard on *hours and human capital* rather than headcounts — one thinks of Hall–Jones-style adjustments for schooling, and of the Penn World Table's move to hours-based labour input. Every quality dimension of labour that goes unmeasured — health, schooling, experience, hours — is pushed wholesale into the residual with an elasticity of one. A “TFP difference” between two countries is a statement about what the analyst failed to measure at least as much as a statement about technology.

Companion code

Every numerical result above is reproduced by `Resolucao/kurlat_ch05_codigo/`. Each script checks its analytical formulas against an independent numerical computation and aborts on any mismatch.

File	What it does
<code>ch05_numerico.py</code>	Problem 5.1: the steady state, the two 100-year simulations, the years to 95%, plus a rerun under the exact (non-approximated) law of motion. Problem 5.2: the table of g_y , each row cross-checked against one exact iteration of the capital recursion, and the limit $\delta(1-\alpha)$. Problem 5.3: the steady-state decomposition. Problem 5.4: the closed form $K_t^{\text{EST}} = 2 - (0.9)^t$ against the raw iteration, and $2^{-0.35}$. Problem 5.5: the TFP bias and a proof that long-run output and wages fall by $1/(1+\phi)$. Problem 5.6: the drift of $\ln(r+\delta)$ under each conjecture. Problems 5.7 and 5.8: the three GDP methods, asserted equal; the factor shares; the interest rate; the growth accounting; and a grid search confirming that $s_{gr} = \alpha$. Problem 5.9: an artificial economy in which the four-step estimator recovers A exactly, the bias is exactly x , and the Gotham formula is recovered after converting to Hicks-neutral units.
<code>ch05_figuras.py</code>	The three figures.
<code>estilo_mpl.py</code>	pgf backend, so figure text is typeset by <code>pdflatex</code> with <code>lmodern</code> and <code>cmap</code> and the PDF stays fully copyable.